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Adjustable dumbbell

Abstract

Provided is a novel adjustable dumbbell, including a handle and counter weights arranged on two sides of the handle, where each of the counter weights takes a dumbbell plate group as a weight core and is limited by a positioning frame; meanwhile, through cooperation of a positioning pin, a mounting hole in the positioning frame and a dumbbell plate unit re-positioning hole, the dumbbell plate unit can be increased/reduced through simple plugging and unplugging of the positioning pin, so that the weight of the dumbbell can be increased/reduced. The novel adjustable dumbbell is simple in structure and convenient to operate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority benefit of Chinese patent Application No. 202410224398.0, filed on Feb. 28, 2024, and the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of fitness equipment, and in particular to a novel adjustable dumbbell.

BACKGROUND

(3) A dumbbell, as widely used fitness equipment, is mainly configured to enhance muscle strength training. An adjustable dumbbell is a dumbbell capable of adjusting the weight according to individual requirements.

(4) The working principle of the adjustable dumbbell is to change the weight by adjusting internal locking mechanism, so that users can choose appropriate weight according to their own situation and training objectives. However, most of the adjustable dumbbells in the prior art are complicated in structure, so it is inconvenient to increase/reduce a dumbbell plate unit at any time; furthermore, during use, the dumbbell plate units at two ends of the dumbbell are prone to shake, so the safety is low.

(5) Therefore, how to develop a novel adjustable dumbbell with a simple structure and capable of locking the position of a weight plate stably has become an urgent problem for people.

SUMMARY

(6) In view of this, the present disclosure provides a novel adjustable dumbbell, so as to solve the problems that the adjustable dumbbell in the past has a complicated structure and is inconvenient to increase/reduce the weight, and a dumbbell plate unit is prone to shake.

(7) The technical solution provided by the present disclosure is specially a novel adjustable dumbbell. The dumbbell includes: a handle and two counter weights; the two counter weights are respectively arranged at two ends of the handle, and each of the counter weights includes: a dumbbell plate group, a positioning frame and a positioning pin; the dumbbell plate group includes a plurality of dumbbell plate units which are clamped sequentially, and each of the dumbbell plate units is provided with a positioning hole penetrating through two sides; the positioning frame is sleeved outside the dumbbell plate group and is fixedly connected to the handle, and the positioning frame is provided with a plurality of mounting holes corresponding to the dumbbell plate units in the dumbbell plate group at intervals; and the positioning pin is inserted in any one of the mounting holes in the positioning frame and the positioning hole in the corresponding dumbbell plate unit, and locks the corresponding dumbbell plate unit and all dumbbell plate units located on an inner side of the corresponding dumbbell plate unit in the positioning frame, or locks the corresponding dumbbell plate unit and all dumbbell plate units located on an outer side of the corresponding dumbbell plate unit in the positioning frame.

(8) Preferably, one side of the dumbbell plate unit is provided with a sunken locking groove, the locking groove penetrates through a top end of the dumbbell plate unit, the other side of the dumbbell plate unit is correspondingly provided with a protruded locking block, and the locking block is in fit sliding clamping connection with the locking groove in the adjacent dumbbell plate unit.

(9) Further preferably, one side of the dumbbell plate unit in the dumbbell plate group located on an end part of an outer side is provided with a sunken locking groove, and the other side is a plane.

(10) Further preferably, an end face of the locking groove and an end face of the locking block are swallowtail-shaped.

(11) Further preferably, the positioning frame includes: a first side plate, a second side plate and a connecting rod; and the first side plate and the second side plate are arranged at intervals and in parallel, the first side plate and the second side plate are fixedly connected to the connecting rod, the connecting rod is provided with the mounting holes at intervals, and the mounting holes are located between the first side plate and the second side plate.

(12) Further preferably, the connecting rod in two positioning frames is one link rod, the handle is a hollow sleeve, and the handle is located between the two positioning frames and fixedly sleeved outside the connecting rod.

(13) Further preferably, the center of each dumbbell plate unit is provided with a mounting hole channel penetrating through two sides in a longitudinal direction, and an upper end of the mounting hole channel penetrates through an upper end of the dumbbell plate unit.

(14) Further preferably, the positioning pin includes: a plug-in bolt rod, a plug-in bolt cap and a magnet; the plug-in bolt rod is U-shaped, one end is open, and the other end of the plug-in bolt rod is closed; the plug-in bolt cap is buckled and sleeved at the closed end of the plug-in bolt rod, and an inner side of the plug-in bolt cap is provided with a mounting groove; and the magnet is mounted in the mounting groove of the plug-in bolt cap.

(15) Further preferably, the adjustable dumbbell further includes: a base; the base is configured to bear the handle and the counter weights at two ends, two ends of the base are bearing areas, limiting bosses are arranged on an upper surface of each of the bearing areas at intervals, and the limiting bosses correspond to the dumbbell plate units in the dumbbell plate group; and a bottom surface of each dumbbell plate unit is provided with a limiting groove matched with the limiting boss.

(16) Further preferably, the base is provided with a plug-in bolt placing groove for placing the positioning pin.

(17) The novel adjustable dumbbell provided by the present disclosure includes a handle and counter weights arranged on two sides of the handle, where each of the counter weights takes a dumbbell plate group as a weight core and is limited by a positioning frame, so that the problem that the dumbbell plate group is prone to shake can be effectively solved; meanwhile, through cooperation of a positioning pin, a mounting hole in the positioning frame and a dumbbell plate unit re-positioning hole, the dumbbell plate unit can be increased/reduced through simple plugging and unplugging of the positioning pin, so that the weight of the dumbbell can be increased/reduced. The novel adjustable dumbbell is simple in structure and convenient to operate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The drawings of the specification that constitute a part of the present application are used to provide a further understanding of the present disclosure. The exemplary embodiments of the present disclosure and the description thereof are used to explain the present disclosure, and do not constitute an improper limitation of the present disclosure. In the drawings:

(2) FIG. 1 is a structural schematic diagram of a novel adjustable dumbbell according to an embodiment of the present disclosure;

(3) FIG. 2 is a structural schematic diagram of a counter weight in a novel adjustable dumbbell according to an embodiment of the present disclosure;

(4) FIG. 3 is a schematic diagram of a novel adjustable dumbbell according to an embodiment of the present disclosure, in which a positioning pin is inserted into a first mounting hole on an inner side for taking;

(5) FIG. 4 is a schematic diagram of a novel adjustable dumbbell according to an embodiment of the present disclosure, in which a positioning pin is inserted into a second mounting hole on an inner side for taking;

(6) FIG. 5 is a structural schematic diagram of a dumbbell plate group in a novel adjustable dumbbell according to an embodiment of the present disclosure;

(7) FIG. 6 is a structural schematic diagram of a positioning frame in a novel adjustable dumbbell according to an embodiment of the present disclosure;

(8) FIG. 7 is a structural schematic diagram of a positioning pin in a novel adjustable dumbbell according to an embodiment of the present disclosure; and

(9) FIG. 8 is a structural schematic diagram of a base in a novel adjustable dumbbell according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(10) The present disclosure will be described in detail below with reference to the accompanying drawings and in conjunction with embodiments. Various examples are provided by way of interpretation of the present disclosure and are not intended to limit the present disclosure. Indeed, it will be apparent to those skilled in the art that modifications and variations may be made in the present disclosure without departing from the scope or spirit of the present disclosure. For example, features shown or described as part of one embodiment may be used in another embodiment to produce yet another embodiment. Therefore, it is desirable that the present disclosure includes such modifications and variations falling within the scope of the appended claims and their equivalents.

(11) In the description of the present disclosure, the terms “longitudinal”, “transverse”, “up”, “down”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom” and the like denote orientation or positional relationships based on those shown in the drawings and are intended for ease of description only and not to require that the present disclosure is necessarily constructed and operated in a particular orientation and thus cannot be construed as a limitation to the present disclosure. The terms “connection”, “connect” and “set” used in the present disclosure

should be understood in a broad sense, for example, which may refer to a fixed connection or a detachable connection; which may refer to a direct connection or an indirect connection through intermediate components; which may refer to a wired electrical connection, a radio connection, or a wireless communication signal connection, and the specific meanings of the above terms may be understood by those of ordinary skill in the art according to a specific situation.

(12) One or more examples of the present disclosure are shown in the accompanying drawings. The detailed description uses numeric and letter marks to refer to features in the drawings. Similar or like reference signs in the drawings and descriptions have been used to refer to similar or like parts of the present disclosure. As used herein, the terms “first”, “second”, “third” and the like are used interchangeably to distinguish one member from another and are not intended to denote the location or importance of individual members.

(13) To solve the problems that the adjustable dumbbell in the past has a complicated structure and is inconvenient to increase/reduce the weight, and a dumbbell plate unit is prone to shake, this embodiment provides a novel adjustable dumbbell. As shown in FIG. 1, the dumbbell mainly includes three parts, namely a handle **1** and two counter weights **2**; the two counter weights **2** are respectively arranged at two ends of the handle **1** and fixedly connected to the handle **1**; and the two counter weights **2** have the same structure and adjustable weight.

(14) Referring to FIG. 2, each of the counter weights **2** includes three parts, namely a dumbbell plate group **21**, a positioning frame **22** and a positioning pin **23**, where the dumbbell plate group **21** takes the weight of the counter weights **2** as a core and includes a plurality of dumbbell plate units **211** which are clamped sequentially; each of the dumbbell plate units **211** is provided with a positioning hole **2113** penetrating through two sides; the positioning frame **22** is sleeved outside the dumbbell plate group **21**; and the positioning frame **22** is provided with a plurality of mounting holes **2231** corresponding to the dumbbell plate units **211** in the dumbbell plate group **21** at intervals.

(15) The positioning pin **23** is inserted in any one or any one group of mounting holes **2231** in the positioning frame **22** and the positioning hole **2113** in the corresponding dumbbell plate unit **211**, locks the corresponding dumbbell plate unit **211** and all dumbbell plate units **211** located on an inner side of the corresponding dumbbell plate unit **211** in the positioning frame **22**; or locks the corresponding dumbbell plate unit **211** and all dumbbell plate units **211** located on an outer side of the corresponding dumbbell plate unit **211** in the positioning frame **22**. At this time, when the handle of the dumbbell is held and is taken upwards, the dumbbell plate unit **211** locked in the positioning frame **22** can be taken together through the positioning pin **23**, and the unlocked dumbbell plate unit **211** is placed at the original place. Therefore, by inserting the positioning pin **23** in different mounting holes **2231** in the positioning frame **22**, the number of the taken dumbbell plate units **211** can be adjusted, the weight can be adjusted, and operation is simple. As shown in FIG. 3, when the positioning pin **23** is inserted in the first mounting hole **2231** in the inner side of the positioning frame **22**, only one dumbbell plate unit **211** on the inner side of the dumbbell plate group **21** can be taken. As shown in FIG. 4, in a case that the positioning pin **23** is inserted into the second mounting hole **2231** in the inner side of the positioning frame **22**, two dumbbell plate units **211** on the inner side of the dumbbell plate group **21** can be taken.

(16) According to the novel adjustable dumbbell provided by the above embodiment, due to the brand new structural design of the structure of the counter weights **2**, each dumbbell plate unit **211** is provided with the positioning hole **2113** which is matched with the mounting hole **2231** in the positioning frame **22**; through the plugging and unplugging operation of the positioning pin **23**, different numbers of dumbbell plate units **211** can be locked in the positioning frame **22**, so that the locked number of the dumbbell plate units **211** can be adjusted, that is, when the weight of the counter weights **2** can be increased or reduced and the structure is simplified, the operation is facilitated. When the dumbbell is lifted, the dumbbell plate units **211** locked in the positioning frame **22** is limited by the positioning frame **22**, so the shaking problem during lifting can be

effectively solved, and the safety coefficient can be increased.

(17) Referring to FIG. 5, this embodiment provides a specific structure of a dumbbell plate unit; one side of the dumbbell plate unit **211** is provided with a sunken locking groove **2111**; the locking groove **2111** penetrates through the top end of the dumbbell plate unit **211**; the other side of the dumbbell plate unit **211** is correspondingly provided with a protruded locking block **2112**; and the locking block **2112** is in fit sliding clamping connection with the locking groove **2111** in the adjacent dumbbell plate unit **211**.

(18) Through the cooperation of the locking groove **2111** and the locking block **2112** between the dumbbell plate units, when the positioning pin **23** is inserted into the positioning hole **2113** of any dumbbell plate unit **211**, the dumbbell plate units **211** of the locking groove **2111** of the dumbbell plate unit **211** are locked together through the cooperation between the locking block and the locking groove, the locking block of the dumbbell plate unit **211** slides out of the locking groove of the adjacent dumbbell plate unit **211**, and the locking block and the locking groove are automatically unlocked.

(19) The shape of the dumbbell plate unit **211** can be set freely according to preferences, for example, triangle and trapezoid. Taking the common square as an example in FIG. 5, the dumbbell plate unit **211** may be made of traditional high-density metal, or may be a structure of building a high-density filler in an injection molding shell. The structure of building the high-density filler in the injection molding shell can control the metal material within 15%, and 85% of the material can be replaced with a natural material and a degradable material, so that carbon can be greatly reduced, and the cost and selling price of products can also be reduced.

(20) To make the dumbbell plate unit **211** and the positioning frame **22** fit better, as an improvement of the technical solution, referring to FIG. 1, one side of the dumbbell plate unit **211** of the dumbbell plate group **21** located on an end part of an outer side is provided with a sunken locking groove **2111**, and the other side is a plane; at this time, the positioning pin **23** is inserted into the positioning hole **2113** of the dumbbell plate unit **211**, and the dumbbell plate unit **211** and all dumbbell plate units **211** on the inner side can be locked; and in a case that one side of the dumbbell plate unit **211** of the dumbbell plate group **21** located on the end part of the inner side is provided with the sunken locking groove **2111**, and the other side is a plane; and at this time, the positioning pin **23** is inserted into the positioning hole **2113** of the dumbbell plate unit **211**, and the dumbbell plate unit **211** and all dumbbell plate units **211** on the outer side can be locked.

(21) To improve the firmness of the clamping connection between the dumbbell plate units **211**, as an improvement of the technical solution, the end face of the locking groove **2111** and the end face of the locking block **2112** are swallowtail-shaped.

(22) Referring to FIG. 2, the positioning frame **22** mainly includes a first side plate **221**, a second side plate **222** and a connecting rod **223**, where the first side plate **221** and the second side plate **222** are arranged at intervals and in parallel; the first side plate **221** and the second side plate **222** are fixedly connected to the connecting rod **223**; the connecting rod **223** is provided with mounting holes **2231** at intervals; and the mounting holes **2231** are located between the first side plate **221** and the second side plate **222**. The connecting rod **223** in the positioning frames **22** at two ends of the handle may be two independent connecting rods, or may be one connecting rod. In a case that the connecting rod is the two independent connecting rods, two positioning frames **22** are connected by the handle **1** and are fixedly welded on the adjacent side plates in the two positioning frames **22** through the two ends of the handle; in a case that the connecting rod is one link rod, the two positioning frames **22** are connected by the connecting rod, the handle **1** is a hollow sleeve, the handle is located between the two positioning frames **22** and fixedly sleeved outside the connecting rod **223**, and the specific structure is shown in FIG. 6; and when the two positioning frames **22** share one connecting rod, the mounting holes **2231** are formed at two ends of the connecting rod **223** at intervals.

(23) To facilitate the mounting of each dumbbell plate unit, as an improvement of the technical

solution, referring to FIG. 5, the center of each dumbbell plate unit **211** is provided with a mounting hole channel **2114** penetrating through two sides in a longitudinal direction; an upper end of the mounting hole channel **2114** penetrates through an upper end of the dumbbell plate unit **211**; and the mounting hole channel **2114** is mainly used for the connecting rod **223** in the positioning frames **22** to pass through.

(24) Referring to FIG. 7, the positioning pin **23** mainly includes a plug-in bolt rod **231**, a plug-in bolt cap **232** and a magnet **233**, where the plug-in bolt rod **231** is U-shaped, one end is open, and the other end is closed; the plug-in bolt cap **232** includes an upper part and a lower part which are buckled and sleeved at the closed end of the plug-in bolt rod **231**; an inner side of the plug-in bolt cap **232** is provided with a mounting groove **2321**; the magnet **233** is mounted in the mounting groove **2321** of the plug-in bolt cap **232**; after the positioning pin **23** is inserted into the mounting hole **2231** in the positioning frame **22** and the positioning hole in the dumbbell plate unit, a force for promoting the positioning pin **23** to be inserted inward can be generated by a magnetic attraction force of the magnet, so that operation is facilitated; meanwhile, after mounting in place, the magnet is tightly attracted to the dumbbell plate unit, thereby avoiding accidental falling and further improving the safety.

(25) The plug-in bolt rod **231** may adopt a U shape in FIG. 7, or may adopt a traditional straight line. The main objective of using the U shape is to improve the stability. The U-shaped plug-in bolt rod is equivalent to using two spaced points to limit the dumbbell plate unit to avoid shaking.

(26) In cooperation with the use of the U-shaped plug-in bolt rod **231**, each dumbbell plate unit **211** is provided with two positioning holes **2113** at intervals; the positioning holes may be formed in the dumbbell plate unit **211** to penetrate through two sides; however, for the processing convenience, preferably, as shown in FIG. 5, two side surfaces of the dumbbell plate unit **211** are provided with sunken arc-shaped grooves as two spaced positioning holes.

(27) Referring to FIG. 8, the adjustable dumbbell further includes: a base **3**; referring to FIG. 1, the base **3** is configured to bear the handle **1** and the counter weights **2** at two ends; two ends of the base **3** are bearing areas **31**; an upper surface of each of the bearing areas is provided with limiting bosses **311** at intervals; the limiting bosses **3113** correspond to the dumbbell plate units **211** in the dumbbell plate group **21**; and referring to FIG. 5, a bottom surface of each dumbbell plate unit **211** is provided with a limiting groove **2115** matched with the limiting boss **311**. When the dumbbell plate group **21** is placed on the base **3**, the limiting groove **2115** on the bottom surface of each dumbbell plate unit **211** cooperates with the corresponding limiting boss **311** to limit each dumbbell plate unit **211**.

(28) Preferably, the base **3** is provided with a plug-in bolt placing groove **32** for placing the positioning pin **23**.

(29) To further improve the safety, a surface of the handle **1** of the dumbbell is provided with skidproof stripes.

(30) The foregoing is merely a preferred embodiment of the present disclosure and is not intended to limit the present disclosure which may be subject to various modifications and variations to those skilled in the art. Any modification, equivalent replacement, improvement, etc. made within the spirit and principles of the present disclosure should be included in the scope of protection of the present disclosure.

Claims

1. A novel adjustable dumbbell, comprising a handle and two counter weights, wherein the two counter weights are respectively arranged at two ends of the handle, and each of the counter weights comprises: a dumbbell plate group, a positioning frame and a positioning pin; wherein the dumbbell plate group comprises a plurality of dumbbell plate units which are clamped sequentially, and each of the dumbbell plate units is provided with a positioning hole penetrating through two

- sides; wherein the positioning frame is sleeved outside the dumbbell plate group and is fixedly connected to the handle, and the positioning frame is provided with a plurality of mounting holes corresponding to the dumbbell plate units in the dumbbell plate group at intervals; and wherein the positioning pin is inserted in any one of the mounting holes in the positioning frame and the positioning hole in the corresponding dumbbell plate unit, and locks the corresponding dumbbell plate unit and all dumbbell plate units located on an inner side of the corresponding dumbbell plate unit in the positioning frame, or locks the corresponding dumbbell plate unit and all dumbbell plate units located on an outer side of the corresponding dumbbell plate unit in the positioning frame.
2. The novel adjustable dumbbell according to claim 1, wherein one side of each dumbbell plate unit is provided with a sunken locking groove, the locking groove penetrates through a top end of the dumbbell plate unit, the other side of the dumbbell plate unit is correspondingly provided with a protruded locking block, and the locking block is in fit sliding clamping connection with the locking groove in an adjacent dumbbell plate unit.
3. The novel adjustable dumbbell according to claim 2, wherein one side of a dumbbell plate unit in the dumbbell plate group located on an end part of an outer side is provided with a sunken locking groove, and the other side is a plane.
4. The novel adjustable dumbbell according to claim 2, wherein an end face of the locking groove and an end face of the locking block are swallowtail-shaped.
5. The novel adjustable dumbbell according to claim 1, wherein the positioning frame comprises: a first side plate, a second side plate and a connecting rod; and wherein the first side plate and the second side plate are arranged at intervals and in parallel, the first side plate and the second side plate are fixedly connected to the connecting rod, the connecting rod is provided with the mounting holes at intervals, and the mounting holes are located between the first side plate and the second side plate.
6. The novel adjustable dumbbell according to claim 5, wherein the connecting rod is one link rod, the handle is a hollow sleeve, and the handle is arranged between two positioning frames and fixedly sleeved outside the connecting rod.
7. The novel adjustable dumbbell according to claim 6, wherein a center of each dumbbell plate unit is provided with a mounting hole channel penetrating through two sides in a longitudinal direction, and an upper end of the mounting hole channel penetrates through an upper end of the dumbbell plate unit.
8. The novel adjustable dumbbell according to claim 1, wherein the positioning pin comprises: a plug-in bolt rod, a plug-in bolt cap and a magnet; wherein the plug-in bolt rod is U-shaped, one end is open, and the other end of the plug-in bolt rod is closed; wherein the plug-in bolt cap is buckled and sleeved at the closed end of the plug-in bolt rod, and an inner side of the plug-in bolt cap is provided with a mounting groove; and wherein the magnet is mounted in the mounting groove of the plug-in bolt cap.
9. The novel adjustable dumbbell according to claim 1, further comprising: a base, wherein the base is configured to bear the handle and the counter weights at two ends, two ends of the base are bearing areas, limiting bosses are arranged on an upper surface of each of the bearing areas at intervals, and the limiting bosses correspond to the dumbbell plate units in the dumbbell plate group; and wherein a bottom surface of each dumbbell plate unit is provided with a limiting groove matched with the limiting boss.
10. The novel adjustable dumbbell according to claim 9, wherein the base is provided with a plug-in bolt placing groove for placing the positioning pin.
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